

# Omron Electronic Thermometer MC 246

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## 1. Introduction

**Brand Name:** Omron Electronic Thermometer MC 246

**Manufactured by:** Omron Dalian Co. Ltd

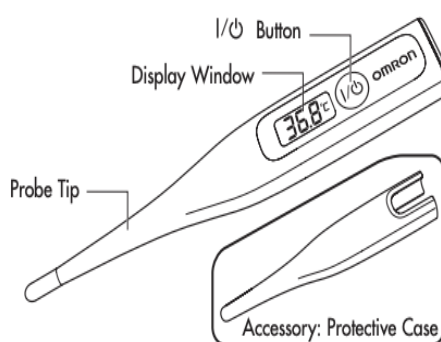
**Imported and marketed by:** Omron Healthcare India Pvt.Ltd.

Core body temperature is an important physiological indicator for self-health assessment and medical diagnosis. A sudden rise in body temperature may indicate a fever, a common symptom in infections. Thermometers are specialized instruments that measure the body's core temperature accurately. Mercury thermometers were once thought to be the gold standard for precisely determining body temperature, but due to the toxic effects of mercury from broken thermometers, as well as the harm broken glass could cause to patients and the health workers, mercury thermometers were replaced by digital thermometers.

Omron Electronic Thermometer MC 246 helps in measuring body temperature by eliminating any broken glass or mercury hazards. It can easily switch between °C and °F, and the device is also water-resistant.

## 2. Description and Method of Use

Omron Electronic Thermometer MC 246 provides a safe, precise, and quick temperature reading. It can be used to measure temperature via oral, rectal, or underarm sites.



### Taking the temperature

- The on/off button should be pressed.  
The display will show a default temperature of 37.0°C (98.6°F).
- Apply the thermometer to the measurement site.  
Use the oral, rectal, or underarm site for measurement.
- Once the reading is ready, the thermometer will emit three beeps. Remove the thermometer from the site and note the result.
- Switch off the thermometer and store it in the protective case.

#### NOTES:

- The results will be automatically stored in the memory.
- Even though the thermometer switches off automatically 30 minutes after use or after 3 minutes if not used, it should be manually turned off to preserve battery life.

## Displaying the temperature

Press the button to display the most recently recorded temperature.

## Correct measurement

### ➤ Oral Use (for ages 4 and older):

- The mouth should be kept closed for at least 5 minutes before attempting a reading.
- Place the thermometer in the mouth under the tongue so that it rests to the left or right of the root of the tongue.
- Use downward tongue pressure to hold the thermometer in place.
- Hold the thermometer to keep it from sliding around in the mouth.

### ➤ Rectal Use:

- Always use a probe cover when taking rectal measurements.
- Used commonly for young children when it is difficult to take an oral or underarm temperature.
- The probe tip should be lubricated with a water-soluble gel. Do not use petroleum jelly.
- Insert the probe gently into the rectum. Never force the probe tip into the rectum if any resistance is encountered.
- Disinfect the thermometer after use.

### ➤ Underarm Use:

- Place the probe tip in the center of the underarm. [The angle should be 35- 45 degrees in relation to the arm. Thermometer Angle].
- Lock the probe tip in the underarm, using the arm to apply slight inward pressure. In the case of infants and very young children, gently hold the arm to prevent movement.

## Error display

| Indicator             | Cause                                                                                 | Remedy                                                                |
|-----------------------|---------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| I                     | Displayed when the temperature of the sensing section is less than 32°C (89.6°F).     | This is not an error, so it may be ignored and the temperature taken. |
| h                     | Displayed when the temperature of the sensing section is greater than 42°C (107.6°F). | Cool the sensing section and then take the temperature.               |
| Temperature displayed | Appears when the surrounding temperature is exceeding 32°C (89.6°F).                  | This is not an error, so the temperature can be taken as it is.       |

## Troubleshooting

| Problem                                                                     | Cause                                                                                                                                                                                                                                     | Correction                                                                                                                                                             |
|-----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Wide differences in readings                                                | The temperature sensing part of the thermometer is being placed at a different location for each reading. • The thermometer is moved while the temperature is being taken. • The mouth is kept open while the measurement is in progress. | Check to see that the thermometer is being placed at the location as described in the “CORRECT MEASUREMENT” section.                                                   |
| Nothing appears in the display section when the power switch is pressed     | The polarity is reversed.                                                                                                                                                                                                                 | Take out the battery and replace using the correct polarity.                                                                                                           |
|                                                                             | The battery is depleted                                                                                                                                                                                                                   | Check to see if  appears in the display section. Replace with a new battery (LR41). |
| 37.0°C (98.6°F) is not displayed during the measurement preparation period. | A temperature between 36.9°C (98.4°F) to 37.1°C (98.8°F) is shown.                                                                                                                                                                        | The thermometer can be used as it is.                                                                                                                                  |
|                                                                             | A temperature other than the range of 36.9°C (98.4°F) to 37.1°C (98.8°F) is displayed, even after the thermometer is switched on and off several times.                                                                                   | Contact your nearest authorized OMRON dealer.                                                                                                                          |

### 3. Principle of Working

The digital thermometer is based on the presence of a sensing part containing a thermistor that senses a physical change (usually electrical resistance) when in contact with the body measurement site and converts that measurement into a temperature. A thermistor is a semiconductor temperature sensor whose electrical resistance decreases on heating and increases on cooling. The thermometer has a tiny battery that sends a current through a wire towards the tip where the thermistor is present. The current from the thermistor is translated into a temperature reading that is displayed on a digital readout.

### 4. Indications

- Measuring body temperature.
- Detection of fever and hypothermia.
- Monitoring progression of illness.
- Monitoring effects of antimicrobial therapy and observing for signs of transfusion reaction.

### 5. Advantages and Disadvantages

#### Advantages

- Rapid temperature measurement.
- Accuracy: Digital thermometers provide more precise readings.

- Convenience and Ease of use.
- No potential health hazard compared to mercury-glass thermometers.
- Safe: reduces the danger of broken glass.

### Disadvantages

- Cost
- Dependency on battery power.
- Inaccurate in case of improper placement techniques.

## 6. Contraindications

### Oral thermometry

- Young children and infants.
- Confused, unconscious patients.
- Patients who have undergone recent oral surgery
- Patient who cannot hold the thermometer in the mouth or have a tendency to bite the thermometer.
- Trauma to the face or mouth
- Avoid monitoring the temperature immediately after smoking, eating, drinking, bathing, and exercising may affect the accuracy.

### Rectal thermometry

- Patients who have undergone recent rectal surgery.
- Patients with rectal pathology.
- Patients with diarrhea.
- Patients having difficulty maintaining the correct position for measurement.

### Underarm thermometry

- Heavy underarm perspiration.
- Assessing the temperature immediately after being under the blanket for a long time.

## 7. Care and maintenance

- Keep the thermometer clean.
- Never use strong jets of water to clean the thermometer, or leave it immersed in water for long periods of time.
- The thermometer should be kept clean with a soft, clean cloth.
- When using alcohol to clean the thermometer, make sure that it does not come in contact with the display.

- For heavy stains, wipe the thermometer with a cloth that has been dampened with water or a neutral detergent solution and then wring the cloth thoroughly. Complete by wiping with a soft dry cloth.
- Avoid the following to prevent damage to the thermometer.
  - Never use benzene, thinner, gasoline or other strong solvents to clean the thermometer.
  - Do not soak the sensing section in alcohol for long periods of time or attempt to sterilize it using hot water (water at a temperature of 50°C (122°F) or higher).
  - Never use ultrasonic washing to clean the thermometer.

### Store the thermometer in the protective case.

Do not store the thermometer in the following types of places.

- Wet locations.
- Locations with high heat and humidity or exposure to direct sunlight. Areas close to heating equipment, dusty locations, or environments where there are high salt concentrations in the air.
- Locations where the unit will be subjected to leaning, shocks or vibrations - Pharmaceutical storage areas or locations where corrosive gases are present.

**Important safety notes:** To assure the correct use of the product basic safety, measures should always be followed including the precautions listed below

#### ➤ **Warning**

- A high or prolonged fever requires medical attention, especially for young children. Please contact your doctor.
- Read and follow carefully the enclosed instructions to ensure accurate temperature readings. Note that temperature readings are affected by several factors including physical exertion, drinking of hot or cold beverages before measurement, as well as the measuring technique.
- Please remain still during the measurement.
- The use of temperature readings for self-diagnosis is dangerous. Consult with your doctor for the interpretation of results. Self-diagnosis may lead to the worsening of existing disease conditions.
- Store the thermometer out of the reach of children. Do not allow children to take their temperatures unattended. Children may injure themselves while attempting to take temperatures without any adult supervision.
- Taking measurements when the thermometer is wet may result in inaccurate readings.

#### ➤ **Caution**

- Never share the thermometer among individuals.
- Do not attempt to incinerate the battery. It may burst.
- Pay attention to polarity (+ –) when replacing the battery. Failure to do so may lead to leakage of fluid, heat generation, or bursting, and finally result in damage to the unit.
- Remove the battery when the thermometer will not be used for 3 months or more. Failure to do so may lead to leakage of fluid, heat generation or bursting, resulting in damage to the unit.
- Never use the thermometer in places where strong static electricity or electromagnetic fields are present, as this may lead to inaccurate readings and may contribute to instrument failure.
- Never force the thermometer into the rectum. Stop insertion and abort the measurement when pain is present. Failure to do so may lead to injury.
- Never attempt rectal measurements on persons with rectal disorders, as it may aggravate or worsen the condition.
- Never step on the unit or the hard case.
- Failure to use a probe cover may lead to bacterial and viral infection.

## 8. Summary

Assessment of body temperature is a core component of vital sign monitoring. It reflects the body's ability to maintain thermal balance and is also an early indicator of infection, inflammation, or systemic dysfunction. A good thermometer should be able to accurately measure the body temperature of people of all ages, be simple and comfortable to use, provide rapid results, not cause cross-infection among patients, and should be not influenced by room temperature. It should be safe and economical. For years, mercury thermometers were the accepted reference method. However, due to concerns over the possible health risk associated with mercury toxicity, the digital thermometer became more favored. Digital thermometers are non-invasive medical devices that can be used in both home and clinical settings.

OMRON Electronic Thermometer MC 246 is a safe, portable, and reliable thermometer that can be used at oral, rectal, and underarm measuring sites. Although the OMRON Electronic Thermometer MC 246 is convenient to use, factors such as smoking, eating, or drinking prior to the measurement, and improper placement technique all contribute to the inaccurate temperature readings. Proper knowledge of correct techniques, storage, and safety measures must be followed to maintain accuracy, reliability, and patient safety.

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